

Machine Learning Approaches in Trojan and Backdoor Detection: A Literature Review

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Abstract—As malware threats are evolving and becoming more complex, especially with backdoor and trojan malwares, have caused massive vulnerabilities in different industries by their nature behaviour of hiding themselves in legitimate software. Thus, with the evolution of the threat landscape, tools that owns the most optimal capability of detecting and mitigating these threats are becoming more crucial. This paper reviews the emerging trends of applying machine learning and deep learning techniques in detecting and mitigating trojan and backdoor malwares. This review also deep-dives into the usage of feature extraction techniques that are most commonly used in the detecting and mitigating malwares using Machine Learning models. Finally, the paper discusses the strengths and weaknesses of the current approaches in detecting and mitigating trojan and backdoor malwares, and provides future directions and the importance of continuous development of machine learning to enhance the detection of trojan and backdoor malwares.

I. INTRODUCTION

Nowadays, the picture of cybersecurity landscape has become increasingly unpredictable, with sophisticated malwares such as Trojans and Backdoors has posed a significant challenge to the cybersecurity experts. These malicious softwares exploit the vulnerabilities in different systems, causing bigger damages to the businesses and individuals. Thus, the need for developing detection and mitigation tools is becoming quinessential. This literature review focuses on exploring the current trends in fighting against these malwares using machine learning, honeypots and log processing techniques. Moreover, this paper also examines the employed data features that helps employing these detection and mitigation tools, such as event IDs, relevant metrics, connection counts, and other relevant metrics. The findings of this paper will be summarized in Table I, which summarizes the research questions for this review, which is the current trends of using machine learning in detecting and mitigating trojan and backdoor malwares. By adopting insights from well-respected conferences and journals, this review will provide a comprehensive overview for

further understanding the current trends of using machine learning in detecting and mitigating trojan and backdoor malwares and provide directions for further researching regarding this topic.

TABLE I
 AIMED RESEARCH QUESTIONS

ID	Research Questions
Q1	What are the popular static analysis techniques in Android malware detection?
Q2	How do these primary studies conduct the empirical experiments of Android malware detection using static analysis?

II. APPROACH TO THE CURRENT THREATS

A. Machine Learning for Malware Detection

The use of machine learning (ML) techniques for malware detection has evolved significantly, especially in the context of backdoor malware attacks. StealthMiner, a machine learning framework that specialized in time series approach, has been created to detect malware embedded with benign applications using Hardware Performance Counters (HPCs). As a result, this method has achieved a remarkable 94% detection rate, outperforming the old Hardware-Assisted Malware Detection method up to 42% [1].

Thus, as old signature-based antivirus solutions could fail to identify embedded malware are crucial, which could makes the virus undetected for a period, the born of StealthMiner to detect Trojan and Backdoor virus is considered crucial (Sayadi et al., 2023).

In contrast, the research on backdoor attacks has pointed out the vulnerabilites of adversarial manipulations in Deep Neural Networks (DNNs). For instance, The Scapegoat Backdoor Attack (SGBA) method introduces triggering benign to mask the real malicious trigger, thus increase the difficulty for detection systems to identify the backdoor. [2]

B. Backdoor Attack Techniques

In recent years, backdoor attacks have gradually evolved, with multiple techniques being employed to enhance its natural behavior of stealthiness. The Scrapegoat Backdoor Attack (SGBA) method has been shown to fade away the detection of all current models [2]. Since traditional methods are becoming obsolete with the modern attack methods, thus this attack method highlights the ongoing fight between attackers and defenders.

In addition, the B3 attack methods also demonstrates the power of black-box backdoor attacks, which adversaries could manipulate models without the need of accessing the model datasets or architectures [3]. This shows the urgent need for a modern and robust detection architectures that could detect both backdoor and embedded malware. However, with the rise of Topological Data Analysis (TDA), it helps detecting data poisoning attacks for the Network Intrusion Detection Systems (NIDS), thus increasing the robustness of the detection systems by detecting the suspicious points in data distributions [4].

C. Challenges in Detecting Backdoor Attacks

Although detection techniques have massively evolved through years, challenges remain in identifying backdoor attacks effectively. The complexity of ML models, specifically Deep Neural Networks (DNNs), makes it hard to differentiate between malicious behaviors and benign. The overparameterization of DNNs could be an ideal place for backdoors to hide themselves [5]. Moreover, with the nature of only focus on raw telemetry data, traditional detection methods often fails to account for the business logic and operational context of ML systems [6].

The nature of backdoor attacks, which is stealthiness, act as a significant challenge for detection systems. Many DL methods, such as Neural Cleanse and Activation Clustering, fail to detect backdoors in big models that being trained under adversarial conditions [7]. Thus, the demand for a breakthrough detection methods that could catch-up with the current threat landscape and effectively identify backdoor malwares without affected the performance of benign inputs, is mandatory.

D. Hardware Security and Backdoor Detection

The application of Machine Learning to detect hardware Trojans and backdoor attacks, has risen as a critical research topic. Machine Learning has been used for incrementing the hardware Trojans detection capabilities, which could be integrated into circuits in the manufacturing phase [8]. These applications have leveraged multiple features, such as identifying suspicious activities and

malicious modifications based on the power consumption patterns and signal analysis.

On the other hand, integrating hardware performance counter has achieved positive results in detecting embedded malware [1]. By inspecting microarchitectural features, cybersecurity experts can develop lightweight detection methods that is suitable for operating in real-time condition, reducing the challenges posed by backdoor malware. This approach highlights the importance of combining statistics from hardware security and machine learning to develop more stable detection systems.

E. Future Directions and Recommendations

As cyber threats, especially backdoor and trojan malware, continues to evolve, it is essential for researchers and cybersecurity experts to remain adaptive and continuously evolve. Future research should focus on improving comprehensive frameworks that could effectively detects malware and backdoor attacks, taking advantage in integrating hardware security and machine learning. Moreover, a standardized rubrics to evaluate the effectiveness of detection frameworks across various scenarios is a must.

Q1: Current trends in detecting and mitigating cyber threats mostly focus on machine learning, anomaly detection, and hardware security techniques. Although machine learning has been proof for detecting embedded malware within benign application, backdoors malware has also witnessed a new method, which trigger the benign applications to mask malicious intent, thus increasing the detecting complication. Hardware security methods also gain some achievements, mostly for identifying backdoor and trojan malwares embedded in hardware. Moreover, anomaly detection is still a critical aspect to consider, since it is used for detecting and analyzing unusual activities of malwares. To sum up, these trends has shown a diverse range of applications to enhancing protection from complicated attacks.

III. THE USE OF MACHINE LEARNING IN CYBER THREAT DETECTION

A. Event IDs and Connection Metrics

Event ID is an essential component in cyber threat detection frameworks. These identifiers are used for identify specific events, such as file access or user login in both applications and operating systems. Combine Machine Learning with Event ID will enable detecting malicious example, such as brute-force attacks in Windows [6].

In addition, connection metrics, including the duration and number of connections, also plays an essential

role in threat detection. These metrics give specific details of network traffic behavior, allow detecting suspicious network-based activities. For example, a significant amount of network traffic from a single source of IP could potentially be a Distributed Denial of Service (DDoS) attack [6]. By using Machine Learning to analyze these metrics, frameworks can better determine between normal and malicious behaviour.

B. Anomaly Detection and Backdoor Attack Detection

Anomaly detection is an indispensable tie of malware detection, which combine multiple features to identify abnormal behaviors. Techniques such as topological data analysis (TDA) are created to utilize capturing unusual numbers in data. In network intrusion detection system (NIDS), TDA can detect abnormal behaviors within network traffic datasets, thus detecting the potential attacks faster [4].

On the other hand, the impressive growth of backdoor attacks has led to the urgent need for creating innovative detection mechanisms. The Scapegoat Backdoor Attack (SGBA) introduces triggering benign applications to hide the malicious malwares, highlights the adaptability of current detection frameworks [2]. In addition, the black-box attacks exploit accessing models using APIs, showcasing the capability of ML-based detection systems [7].

C. Microarchitectural and Opcode Analysis

Microarchitectural infrastructures, specifically captured by Hardware Performance Counters (HPCs), have shown its effectiveness in finding embedded malware within benign applications. It utilizes instructions and other HPCs methods to detect malware at runtime, which reached an impressive detection rate of 94% [1]. Thus highlights the importance of specific Microarchitectural features in detection performance.

Moreover, the analysing of opcode features, which refers to analyse the operation at the instruction level of a program, also has proofed to achieved high capabilities of detecting threats. Koylu et al. (2023) has indicated that since opcodes often co-existed in malicious applications, allowing ML models to achieve high recall rates and stable precision in identifying Android malware [8].

D. Further Achievements in Threat Detection

The application of Machine Learning in detecting malwares has witnessed some remarkable achievements. The integration of Machine Learning of Machine Learning models to detect different types of malwares, including backdoors and trojans, has significantly enhanced the detection accuracy [1]. Sayadi et al. (2023) has also demonstrates the utilization of minimal features, such as

branch instructions, while still retaining high detection rates.

In addition, the creation of ensemble models that integrate anomaly detection to traditional machine learning has demonstrates some notable efforts in addressing imbalanced datasets, specifically in classifying Android malwares [9]. With the models have exceed an 0.98 F1-scores in multiple datasets, highlighting the accurate capability of classifying malicious applications [9].

Moreover, the non-stop growth of Machine Learning has also led to the broader application of defense frameworks against adversarial attacks. The introduction of AEGIS has significantly addressed to the backdoor attacks in big models, guarding the security of software that integrated Machine Learning [7].

Q2: The application of machine learning for detecting cyber threats has not only proof the efficiency and accuracy in identifying malicious activities but has also assisted the future development of innovative methods to fight against the advanced threats. Researches and improvements in this topic continue to open a promise world for more robust and resilient systems, establishing the potential of using machine learning in cybersecurity.

IV. DISCUSSION

This literature review on the integration of Machine Learning (ML) in detecting backdoors and trojans reveals a promising yet challenging picture. The advance of Machine Learning, specifically in the context of Hardware-Assisted Malware Detection (HMD) and anomaly detection, has highlighted a potential enhancement of cybersecurity measures. For example, the StealthMiner approach that helps capturing microarchitectural features using Hardware Performance Counters (HPCs), has proofed a notable embedded malware detection rate of 94% [1]. Thus showcases that exclusive Machine Learning models could detect backdoor threats better than traditional methods.

On the other hand, the integration capabilities of these methods still have limitations. As mentioned in this review, the dataset must be carefully cleaned to ensure its integrity and relevance, such as the Topological Data Analysis (TDA) needs to have a thorough network traffic data processing in order to achieved meaningful features and higher detecting rate [4]. Thus, while Machine Learning is really efficient in detecting malicious activities, it may requires some quality pre-processing steps, which is not suitable in the real-world scenarios where time and resources are limited.

In addition, the focus of many studies seems to be narrow. Ying He et al. (2024) has pointed out that most of the studies only focus on specific type of

threats and dataset, which could significantly deflects the performance of detection based on the attack type and model architecture [2]. This raises concerns about the dynamic capability of Machine Learning in finding across different threat landscapes. For example, certain Machine Learning models could excel in detecting a specific malware, they could also provide false result in detecting other types of malwares, especially for complex adversarial machine learning techniques [3].

Moreover, the usage of computational resources and the complexity of model training exploit some challenges for companies with limited resources. Bitton et al. (2023) has pointed out that many Machine Learning systems are ready to use "out of the box" and require careful setup procedures to fit with the required contexts and threats [6]. This requires a careful consideration from organizations in which case to use this model, as well as the further adjustments and trainings to catch up with the future threats.

To sum up, while this literature review showcases the application of machine learning in trojans and backdoors detection, the real-world integration of these methods remain complex. The need for high quality datasets, sophisticated preprocessing procedures, and the narrow use cases suggests that organizations must understand the limitations and requirements of these methods. As the field continues to develop, further researches are necessitate to fix the adaptability and increase the robustness of Machine Learning solutions in order to apply it in wider cybersecurity contexts.

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